

ARRAYS

Q) Convert Arabic To Martian Numerals

DS	Session	Arrays	Question Information	Level 1 Challenge 21
Problem Description: How many Y's did a Roman Centurion make a day in cold hard Lira? About a C's worth! Turns out, Martians gave Rome the idea for their number system. Use the conversion charts below to help translate some Martian numbers! Note, that unlike the Roman Numerals, Martian Numerals reuse symbols to mean different values. B can either mean '1' or '100' depending on where it appears in the number sequence. Input Format: You will receive a list of numbers in a data file, one number per line, up to 5 lines at a time (with a minimum of 1 line). No number will exceed 1000, or be less than 1. Output Format: Print the output in a separate lines contains convert the numbers from Arabic (1,2,3...10...500...1000) to Martian (B,BB,BBB...Z...G...R) numerals.				

CODE:

```
EXPLORER
  TEST
    C main.c
    F main.exe

C main.c
1  #include <stdio.h>
2  #include <string.h>
3
4  const char *thousands[] = {
5      "", "B"
6  };
7
8  const char *hundreds[] = {
9      "", "B", "BB", "BBB", "G", "Z", "ZB", "ZBB", "ZBBB", "BR"
10 };
11
12 const char *tens[] = {
13     "", "Y", "YY", "YYY", "G", "L", "LY", "LYY", "LYYY", "YB"
14 };
15
16 const char *ones[] = {
17     "", "B", "BB", "BBB", "Z", "G", "GB", "GBB", "GBBB", "BZ"
18 };
19
20 void arabicToMartian(int num, char *result) {
21     result[0] = '\0';
22
23     strcat(result, thousands[num / 1000]);
24     num %= 1000;
25
26     strcat(result, hundreds[num / 100]);
27     num %= 100;
28
29     strcat(result, tens[num / 10]);
30     num %= 10;
31
32     strcat(result, ones[num]);
33 }
34
35 int main(void) {
36     int num;
37     char martian[64];
38     int linesRead = 0;
39
40     while (linesRead < 5 && scanf("%d", &num) == 1) {
41         if (num >= 1 && num <= 1000) {
42             arabicToMartian(num, martian);
43             printf("%s\n", martian);
44         }
45         linesRead++;
46     }
47
48     return 0;
49 }
```

OUTPUT:

```
PS C:\Users\CH.SC.U4CSE25255\Projects\test> ./main
1
B
100
B
56
LGB
1000
```